UNESCO x Nestlé Global Youth Grant Scheme

"Youth Impact – Because You Matter: For Environmental Sustainability and Climate Action"

Concept Note

Project Information

Project Title: Oasis
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Geographic Scope / Country of Implementation: Qatar and MENA region

I. Summary Description

The proposed initiative introduces a self-contained, low-power digital learning platform designed to make quality education accessible in regions with limited internet or electricity. Built on Raspberry Pi hardware, the system functions as a local mini-server that allows teachers to upload course materials, presentations, and assessments, which students can access wirelessly within the classroom. This model eliminates reliance on cloud infrastructure, enabling fully offline and energy-efficient learning environments.

The project directly addresses educational inequality by equipping teachers in underserved schools with the tools to deliver interactive, technology-based lessons regardless of connectivity barriers. It also contributes to global climate goals by reducing the energy footprint of digital learning systems. Implementation begins in Qatar, where the prototype will be developed, tested, and refined before being expanded to other low-resource regions.

The initiative aligns with SDG 4 (Quality Education), SDG 9 (Industry, Innovation and Infrastructure), and SDG 13 (Climate Action). Over the five-month implementation period, the project will train educators, deploy offline learning devices, and establish a replicable model for sustainable, inclusive, and climate-conscious education.

II. Overall Purpose and Relevance

State the purpose of the project and expected impact to which it aims to contribute. Including needs, issues addressed and baselines:

State how the project will directly contribute to change people's lives in the long term:

This project addresses a structural barrier to human development: millions of children and young people are excluded from quality education not because of ability or motivation, but because their schools and communities lack the basic infrastructure that modern learning assumes – stable electricity, reliable internet access, and affordable digital tools. According to UNICEF and the International Telecommunication Union, **two-thirds of the world's school-age children – about 1.3 billion learners aged 3 to 17 – do not have internet access at home**, which means they cannot benefit from most digital learning platforms currently in use [1]. In parallel, access to even the most basic school infrastructure remains uneven: UNICEF notes that, during pandemic, **over 200 million children attend primary schools with no electricity**, and in Sub-Saharan Africa specifically, **only about 31% of primary schools have electricity access**, making consistent delivery of digital education almost impossible [2]. These gaps directly reproduce inequality between urban and rural

areas, between richer and poorer communities, and often between boys and girls, because those who cannot access learning tools fall behind academically, socially, and economically.

The purpose of this project is to provide a practical, low-barrier way to deliver high-quality learning in precisely these low-resource, high-exclusion environments. The core intervention is a self-contained, low-power local learning hub built on a Raspberry Pi–class device that acts as an offline mini-server. A teacher can load course materials – presentations, readings, exercises, and quizzes – onto the device and bring it physically into the classroom. Students then connect locally to access lessons and complete assessments without needing an external internet connection or a full computer lab. Because the unit is extremely energy-efficient and capable of running on limited or intermittent power, it can operate in classrooms that do not have reliable grid electricity and would otherwise be shut out of digital education. In other words, the project is not trying to make disadvantaged schools copy high-bandwidth, always-online systems; it is redefining "digital access" around the actual conditions on the ground. This directly targets the baseline problem identified above: children in communities with little or no infrastructure are currently excluded from digital learning and therefore from many of the skills, literacies, and opportunities that education is supposed to unlock [1].

The project is also designed to respond to the climate and energy dimension of education access. Conventional e-learning assumes constant connectivity to cloud platforms and data centers. Maintaining and serving that content consumes electricity at multiple layers – device use, data transmission, and remote compute – and that electricity is still largely generated from fossil fuels. World Bank data show that in "least developed countries," nearly **half (about 49.8%) of total electricity generation still comes from fossil fuel sources such as oil, gas, and coal**, which means that every marginal increase in energy demand tends to carry a high carbon cost [3]. At the global level, coal alone remains responsible for roughly one-third of electricity generation, and despite gradual decline, it is still the single largest source of power [4]. By delivering curriculum, assessment, and guided self-study locally on a very low-power device, this project aims to expand learning without proportionally expanding energy consumption and associated emissions. In practical terms, it treats access to education and climate responsibility as linked challenges rather than separate agendas: it is an attempt to prove that it is possible to reach excluded learners without forcing schools to depend on high-carbon digital infrastructure that they cannot afford and that the planet cannot sustain.

In the long term, the expected impact is twofold. First, the project creates immediate educational continuity for learners in fragile, underserved, or infrastructure-poor contexts. When students can regularly access structured lessons and complete quizzes – even in areas with weak electricity and no internet – they are less likely to fall behind, more likely to build foundational skills in literacy, numeracy, science, and digital navigation, and more able to progress toward secondary and post-secondary opportunities. This directly affects life trajectories: education is closely tied to later employability, income security, and the ability to participate in civic and economic life. Second, the project cultivates a generation of students and teachers who are both digitally capable and climate-aware. Because the platform is intentionally energy-efficient and incorporates sustainability-related learning content, it normalizes the idea that essential services like education should be delivered in ways that respect environmental limits. Over time, this equips young people in marginalized communities not only with knowledge, but with the mindset and practical exposure to build solutions in their own contexts – increasing their chances of dignified work, local leadership, and long-term resilience.

III. Implementation Strategy

Alignment with Sustainable Development Goals (SDGs; link), UNESCO priorities (link), and Nestlé's youth empowerment goals (link)

The project's implementation strategy focuses on building a **self-contained, low-power learning platform** that enables teachers to deliver digital education even in areas without stable electricity or internet connectivity. The device functions as a **local mini-server**, allowing educators to upload their full courses — presentations, readings, assessments, and quizzes – which students can then access wirelessly within the classroom through any smart device. It operates independently from the cloud, making it possible to conduct interactive lessons entirely offline. An integrated **AI-based chatbot** assists both teachers and students: for teachers, it helps organize content and quizzes; for students, it supports learning by answering questions, explaining topics, and guiding self-study. This design transforms the device into an all-in-one educational ecosystem that ensures continuity of learning even when digital infrastructure is unavailable or unreliable.

Implementation begins with capacity building for teachers rather than students. Each participating school designates educators responsible for maintaining the device and managing the uploaded content. Training sessions will cover platform navigation, content uploading, quiz generation, and troubleshooting. Teachers retain full control over course materials, privacy, and assessment design. This approach preserves academic integrity while providing educators with a ready-to-use digital classroom that can function without internet access. Once operational, the system allows an entire school to run digital courses offline, helping eliminate disruptions caused by poor connectivity or power shortages.

Strategically, the project is aligned with several key Sustainable Development Goals and global frameworks. It directly advances **SDG 4 (Quality Education)** by expanding access to inclusive and equitable learning environments, particularly in disadvantaged or infrastructure-poor regions. The platform ensures that both male and female students have equal access to course materials and interactive learning resources, contributing to targets 4.1, 4.5, and 4.a. Through its energy-efficient design the project supports **SDG 7 (Affordable and Clean Energy)** and **SDG 9 (Industry, Innovation and Infrastructure)** by offering a practical model for low-carbon educational technology that can be locally deployed and maintained.

The project also contributes meaningfully to **SDG 13 (Climate Action)** by demonstrating how the education sector can actively reduce its carbon footprint. Traditional cloud-based learning systems require continuous data transfer and high-energy data centers. By decentralizing content delivery and keeping computation local, the platform reduces electricity consumption. Moreover, the device includes pre-loaded educational modules on sustainability and renewable energy, so that students not only benefit from a low-carbon learning system but also learn **why** such systems matter. This dual effect – emission reduction and climate literacy – reinforces both mitigation and adaptation objectives within SDG 13.

In alignment with UNESCO's *Education for Sustainable Development 2030 Framework* and *Nestlé Needs YOUTH* initiative, the project prioritizes long-term sustainability and practical skill transfer. Teachers gain digital competencies that strengthen their instructional capacity and confidence in integrating technology into classrooms, while students benefit from structured, autonomous learning experiences guided by their educators. The initiative ultimately demonstrates that educational inclusion, technological innovation, and environmental responsibility can advance together – ensuring that the right to quality education is realized sustainably and efficiently across all contexts.

Framework / Policy Area	Relevant Target or Initiative	Project Contribution	Expected Indicator / Outcome
SDG 4 – Quality Education	4.1, 4.5, 4.a	Expands access to digital learning in low-connectivity schools through teacher-managed offline portable mini-servers	% increase in students accessing full courses offline
SDG 7 – Affordable and Clean Energy	7.1, 7.3	Operates with high energy efficiency under standard electrical supply	Energy use per student (kWh) significantly lower than cloud-based learning
SDG 9 – Industry, Innovation & Infrastructure	9.c	Provides small-scale, locally managed digital infrastructure for schools	# of schools operating local servers
SDG 13 – Climate Action	13.3	Reduces energy demand of digital education and promotes efficient resource use	Estimated electricity savings vs. online delivery (kWh)
UNESCO ESD 2030 Framework	Sustainable learning environments & teacher capacity building	Integrates energy-efficient technology into education delivery	# of teachers trained and using the platform
Nestlé Needs YOUth Initiative	Employability & digital skills	Strengthens teachers' digital competence and classroom innovation capacity	# of educators actively using digital materials offline

Stakeholders: Beneficiaries and partners (Identifying key partners and beneficiaries and their roles. For the beneficiaries, differentiating between direct and indirect. Differentiating should also be at least sex and age-disaggregated, and if possible provided with other forms of disaggregation, e.g. profile, geographic location/area, etc.)

The project's implementation model relies on a network of aligned stakeholders who contribute to, benefit from, and sustain the initiative. Its impact is structured through two layers of beneficiaries — **direct** and **indirect (or secondary)** — and supported by a coalition of key partners across the technical, institutional, governmental, and financial levels.

Direct Beneficiaries

The **direct beneficiaries** are those who will directly interact with and utilize the educational platform:

- **Teachers** are the primary users of the device. They upload course materials, manage digital lessons, and assess students through the offline interface. The platform provides them with

an easy, stable digital teaching environment that does not rely on internet connectivity. Training will be provided to help them integrate the tool into existing curricula and to increase their digital competencies.

- **Students** are the end-users of the system. They benefit from continuous access to structured lessons, quizzes, and interactive learning tools. The integrated chatbot supports self-study, answers subject-related questions, and assists in comprehension of course materials. Students in low-resource areas gain access to the same quality of learning experience as peers in more connected environments.
- **Schools** as institutions are also direct beneficiaries. They receive the devices, participate in training, and integrate the system into their classrooms. Schools achieve immediate gains in teaching consistency, digitalization, and reduced dependence on external infrastructure.

Indirect (Secondary) Beneficiaries

The **indirect beneficiaries** include broader social and institutional actors who gain from the project's implementation:

- **Local and national education authorities** benefit from improved data on educational access and the ability to deploy digital learning without costly broadband infrastructure. Their role includes policy support, monitoring, and long-term integration into national education strategies.
- **Parents and local communities** experience social benefits from the increased educational continuity of their children and reduced family costs associated with online access.
- **Society and economy at large** gain from the rise of digitally literate students and teachers, contributing to a stronger workforce prepared for the digital economy and civic participation.

Key Partners and Collaboration Framework

The project's successful delivery depends on a strong collaboration between public, private, and international partners who share a commitment to accessible and sustainable education.

1. **Technical Development Team** – responsible for designing, assembling, and maintaining the Raspberry Pi–based platform. They handle software updates, troubleshooting, and teacher support. Their role ensures the reliability and scalability of the system.
2. **UNESCO Mentorship and Program Alignment** – as a global partner, UNESCO contributes through its *Youth Impact and Education for Sustainable Development* initiatives. This partnership provides technical guidance, visibility, and knowledge exchange. The collaboration ensures that the project remains consistent with UNESCO's global education priorities and SDG implementation strategies.
3. **Government Authorities (Ministries of Education and ICT)** – essential for policy endorsement and large-scale adoption. Integrating the platform into the official curriculum or national ICT strategies will require coordination with these agencies. The shared objective is to make equitable digital learning a public standard, not a private service.
4. **Private Donors and Philanthropic Foundations** – financial partners who support the purchase and distribution of devices to schools. Since all hardware is provided to schools

free of charge, donations are the key funding mechanism. Partnerships with education-focused foundations and CSR (Corporate Social Responsibility) programs will ensure sustainability and scaling beyond the pilot phase.

5. **Academic and Training Institutions** – partner universities or teacher-training centers that will host workshops, provide training to educators, and evaluate pedagogical outcomes.

Each partner has a clear and complementary role — ensuring that the initiative remains technically stable, financially viable, and educationally aligned with national and international goals.

Beneficiaries Overview Table

Category	Group	Description	Expected Benefits
Direct Beneficiaries	Teachers	Educators operating the platform and managing materials	Increased digital capacity and teaching efficiency
	Students	Learners using the system to access offline lessons and quizzes	Equal access to structured education and improved engagement
	Schools	Institutions hosting and sustaining the system	Digitalization of curriculum delivery; improved continuity of lessons
Indirect Beneficiaries	Education Authorities	Local and national government agencies	Reduced infrastructure costs; improved access metrics
	Parents & Communities	Families supporting children's education	Greater educational stability and reduced online costs
	Broader Society	Regional and national communities	Stronger digital literacy and human capital formation

Key Partners Table

Partner Type	Partner Name / Example	Role in the Project	Contribution / Expected Outcome
Technical Partner	Internal Development Team	Design, assembly, and maintenance of the Raspberry Pi platform	Continuous technical reliability and scalability
International Partner	UNESCO Youth Impact Programme	Mentorship, framework alignment, and visibility	Knowledge exchange and compliance with ESD 2030 goals
Governmental Partner	Ministries of Education, ICT, and related agencies	Policy support, pilot approval, and curriculum integration	Institutional adoption and national scalability
Financial Partners	Private donors, philanthropic	Funding for devices and training programmes	Free equipment distribution and long-term sustainability

Partner Type	Partner Name / Example	Role in the Project	Contribution / Expected Outcome
	foundations, CSR programmes		
Academic / Training Institutions	Universities, teacher-training centers	Deliver workshops and track pedagogical impact	Skilled educators and measurable educational outcomes

Results Chain (Indicating how the project is expected to lead to desired results, through key elements of the pathway of change and underlying assumptions that allow for a successful cause-and-effect relationship between the activities and the different levels of results (i.e. output, outcome, impact))

Impact	Expanded access to quality and inclusive digital education for schools in low-connectivity or resource-constrained regions. The project contributes to SDG 4 (Quality Education) by enabling offline learning environments, SDG 9 (Innovation & Infrastructure) through localized digital systems, and SDG 13 (Climate Action) by promoting energy-efficient solutions.

Assumptions	• The Ministry of Education and other government authorities remain committed to integrating digital learning tools in national school curricula. • Donor and private sector support for inclusive education continues throughout the project cycle. • Schools maintain basic infrastructure (electricity, classrooms, and administrative support) necessary to host digital platforms.

Outcome	Teachers and students in targeted schools effectively use the offline digital learning platform to enhance the quality of classroom instruction and student engagement.

Assumptions	Teachers actively participate in training and continue applying acquired digital teaching skills	School administrators allocate time and space for platform-based lessons	The platform remains technically stable and user-friendly for teachers with limited ICT skills	Local education authorities encourage schools to sustain usage beyond the pilot phase

Outputs	Output 1.1 Offline digital learning platform developed and pre-installed on Raspberry Pi devices for deployment in low-connectivity schools	Output 1.2 Teachers' capacities strengthened through training workshops on using and maintaining the platform	Output 1.3 Partnerships established with government authorities and UNESCO-affiliated networks to align content with curriculum standards	Output 1.4 Monitoring and feedback mechanisms established to evaluate platform effectiveness and learning outcomes

Assumptions	Developers and technical partners ensure timely delivery of stable, secure, and compatible software	Teachers attend and complete training sessions and receive institutional support from school leadership	National and local education authorities cooperate in content validation and policy alignment	Schools and teachers provide reliable feedback data, and monitoring tools remain accessible throughout implementation

Activities	1. Develop and test the offline platform prototype using Raspberry Pi hardware	1. Conduct teacher-training workshops in collaboration with local education offices	1. Hold coordination meetings with government and UNESCO partners	1. Design and deploy a simple monitoring dashboard for usage and impact metrics
	2. Translate and adapt open educational resources (OER) into local languages	2. Produce short user manuals and video tutorials	2. Align platform modules with existing national education strategies	2. Conduct quarterly school visits/meetings for observation and interviews
	3. Pre-install the software on distributed devices	3. Provide follow-up mentoring sessions for participating teachers	3. Formalize MoUs with key partner institutions	3. Analyze collected data and publish periodic evaluation briefs
	Quality assurance and monitoring Conduct regular system functionality checks and bug tracking	Quality assurance and monitoring Collect feedback forms and conduct post-training assessments	Quality assurance and monitoring Ensure documentation and approval of all partnership deliverables	Quality assurance and monitoring Maintain continuous data validation and reporting to project steering group

IV. Risk Assessment and Sustainability

Outlining the **main risks** - strategical: environmental, political, economic, social and reputational risks; financial, legal/compliance; governance or operational to successful delivery of this project - and how they would be addressed.

Indicating how the **benefits derived from the project will be maintained over time** and beyond the Grant Scheme support. Highlighting assumptions, conditions, mechanisms, tools and processes to be put in place regarding the continuation, replication and/or scaling-up of activities after funding has ceased.

Risk Category	Description	Impact / Likelihood	Mitigation Strategy
Technical / Operational	Unexpected hardware or software malfunctions during classroom deployment may delay teaching or limit platform usability.	*Medium impact, medium likelihood.*	Conduct quality assurance tests before each deployment; keep 5–10% of spare devices available for replacement; maintain an open communication line with the development team for bug fixes.
Institutional / Coordination	Delayed administrative approval or limited engagement from local education authorities could slow project rollout.	*High impact, medium likelihood.*	Formalize MoUs early with schools and local authorities; engage Ministry representatives during teacher training sessions to ensure institutional ownership.
Financial	Fluctuations in component prices or donor delays could affect the purchasing schedule.	*Medium impact, low likelihood.*	Retain a contingency allocation (Miscellaneous fund, 6% of total budget) to absorb small cost increases; maintain flexible procurement scheduling.
Social / Adoption	Teachers may be hesitant to adopt new technology or integrate it into lesson delivery.	*Medium impact, medium likelihood.*	Provide user-friendly interfaces and continuous support; highlight real classroom benefits during training; use pilot success stories to encourage adoption.
Environmental / Infrastructure	Power instability or insufficient classroom infrastructure in target schools may limit usage time.	*Low impact, medium likelihood.*	Prioritize pilot schools with stable minimum infrastructure; provide backup storage for offline course continuity; schedule sessions during reliable electricity hours.

The project is designed for long-term sustainability across technical, institutional, and community dimensions:

1. **Technical Sustainability:**

   The Raspberry Pi devices operate offline, require minimal maintenance, and rely on open-source software, ensuring independence from costly proprietary systems. All resources are stored locally, enabling updates and usage even in low-connectivity regions.

2. **Institutional Sustainability:**

   Through collaboration with schools and government authorities, the platform will be gradually integrated into existing teaching frameworks. Teachers trained under this project will continue managing and updating course content beyond the pilot period, ensuring long-term adoption.

3. **Community and Capacity Sustainability:**

   The "train-the-trainer" model empowers teachers to later mentor colleagues in their own schools. This cascading approach multiplies the project's reach without additional financial input.

4. **Environmental Sustainability:**

   By eliminating dependence on cloud servers and minimizing power consumption, the project significantly reduces its carbon footprint. The devices consume less than 10 watts per unit, demonstrating an environmentally responsible model for digital education delivery.

V. Project Timeline

Calendar of Activities (starting October 2025):

Month	Description of Activities Planned
October 2025	Activity 1.1 – Development Strategy. Activity 1.2 – Finalize Project Scope.
November 2025	Activity 2.1 – Start development of the back end for the new system. Activity 2.2 – Start working on brand identity. Activity 2.3 – Business Model Strategy.
December 2025	Activity 3.1 – Start development of the front-end for the new system. Activity 3.2 – Start second phase of back-end development.
January 2026	Activity 4.1 – Finalize development. Activity 4.2 – First round of Raspberry-Pi purchases. Activity 4.3 – Design the hardware enclosure of the devices. Activity 4.4 – Finalize physical product design. Activity 4.5 – Start onboarding of beta testers. Activity 4.6 – Collect feedback.
February 2026	Activity 5.1 – Implement Feedback. Activity 5.2 – Second round of Raspberry-Pi purchases. Activity 5.3 – Onboard the second round of beta testers. Activity 5.4 – Develop a large-scale onboarding plan.
March 2026	Activity 6.1 – Create documentation and manuals. Activity 6.2 – Create easy tutorials inside the platform itself. Activity 6.3 – Start exploring potential partnerships, facilitating expansion. Activity 6.4 – Third round of Raspberry-Pi purchases. Activity 6.5 – Onboard the third round of beta testers

V. Monitoring and Reporting

(Indicating data collection methods: For example, through surveys, testimonials, progress reports, impact assessments. Including the engagement of beneficiaries and partners in monitoring.)

A. Monitoring Approach

The project will apply a mixed monitoring framework combining quantitative and qualitative methods:

- **Data collection:** Monthly tracking of device usage, number of active courses, teacher participation rates, and feedback surveys.
- **Frequency:** Internal monitoring monthly; external reporting every two months.
- **Responsible parties:** Project coordination team (technical + field) and designated teacher focal points in each school.
- **Tools:** Excel-based dashboard for data consolidation, feedback forms (digital and printed), and brief qualitative interviews during school visits.
- **Indicators:**
    - Number of operational devices and functioning classrooms using the system.
    - Number of teachers trained and actively uploading materials.
    - Number of students participating in offline assessments.
    - User satisfaction and functionality feedback.

Findings will be compared to the baseline established during the first pilot month to assess improvement trends.

B. Reporting and Learning

Project progress will be reported **bi-monthly** to the UNESCO x Nestlé GYGS coordination team and shared with local education authorities.

Reports will include:

- Key performance indicators aligned with the Results Chain.
- Financial expenditure updates and variance notes.
- Lessons learned and recommendations for scaling.

A short **final report** at the end of the five-month cycle will summarize achievements, challenges, and adjustments made during implementation.

Lessons learned will directly inform the next phase of expansion in collaboration with governmental and donor partners, ensuring that insights gained in Qatar serve as a model for replication in other low-connectivity regions.

VI. Expected Budget Breakdown

Activity	Expected Amount (USD)	Description	Date
Activity 4.2 – First round of Raspberry-Pi purchases	900	First procurement batch for pilot deployment (Raspberry Pi devices, SD cards, power adapters, and basic casings).	Early January
Activity 4.5 – Start onboarding of beta testers	150	Initial onboarding of first beta testers – includes printed guides, small materials, and coordination expenses.	Late January
Activity 5.2 – Second round of Raspberry-Pi purchases	1800	Second batch of devices to expand the pilot phase to additional schools.	Early February
Activity 5.3 – Onboard the second round of beta testers	300	Orientation and training for the second group of testers (materials, logistics, light refreshments).	Mid-February
Activity 6.1 – Create documentation and manuals	150	Preparation of user manuals, quick-start guides, and troubleshooting materials for teachers.	Early March
Activity 6.4 – Third round of Raspberry-Pi purchases	2700	Third procurement batch for final deployment in partner schools (devices, accessories, packaging).	Mid-March
Activity 6.5 – Onboard the third round of beta testers	450	Training and onboarding of the third testing group (distribution of devices and support materials).	Late March
Marketing	1000	Design of visual materials, social-media outreach, and communication with schools and partners.	January – March
Development Costs	500	Final software development and platform integration (interface refinement, testing, plug-ins).	November – January
Miscellaneous	550	Reserve for minor unforeseen costs (transport, printing, spare parts, adapters).	All months

**** Please note that your proposed budget (USD) must cover the entire grant amount you will receive under the Grant Scheme (i.e 8,500 USD).*